

MGMT 5900 – Data Visualization

Fall 2025

Session 052

Mon, Wed 11:30 am – 1:00 pm

CRN# 23593

Course Location: RAWL 3082

Brightspace Course Link: <https://purdue.brightspace.com/d2l/home/1369181>

Session 067

Mon, Wed 2:50 pm – 4:20 pm

CRN# 24625

Course Location: RAWL 1057

Brightspace Course Link: <https://purdue.brightspace.com/d2l/home/1369138>

Course Structure/Modality: LEC 2x per week

Credit Hours: 2

Prerequisites (if any): None

CONTACT INFORMATION & OFFICE HOURS

Instructor: Dr. Cecilia Ying

Office Location: KRAN 534

E-mail: CeciliaYing@purdue.edu

Office Hours: Monday 1:30 pm – 2:00 pm

Note: Email responses are typically between 24-48 hours (usually faster). If, for some reason, you do not have a response from me within 48 hours, please email me again.

COURSE DESCRIPTION

Explore the art and science of data visualization in this 7-week graduate course! You'll learn how to communicate complex data insights effectively through visualization, regardless of the tools you use. We'll concentrate on the key concepts behind creating impactful graphs and charts that tell a story, inform decisions, and captivate audiences. By mastering the principles of data visualization, you'll be able to produce compelling visuals that clearly and powerfully convey your message.

By the end of this course, students should be able to:

1. Understand the principles of effective data visualization for communication
2. Learn how to choose the right graph or chart for your data and message
3. Master techniques for creating clear, impactful, and engaging visualizations
4. Develop skills to iterate and refine visualizations for better communication
5. Apply data visualization principles to real-world data challenges

COURSE REQUIREMENTS

No textbooks required. Online materials will be provided in the course notes.

Optional

- Few, Stephen (2012). *Show Me the Numbers. Designing Tables and Graphs to Enlighten*. (Second Edition). Analytics Press.
- Sleeper, Ryan (2018). *Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master*. 1st edition. Beijing: O'Reilly Media.
- Jones, Ben (2014). *Communicating Data with Tableau: Designing, Developing, and Delivering Data Visualizations*. 1st edition. Sebastopol, CA: O'Reilly Media.
- Yau, Nathan (2013). *Data Points: Visualization That Means Something*. Indianapolis, IN: John Wiley & Sons, Inc.

ASSESSMENTS/ASSIGNMENTS

Evaluation will be based on your understanding of the material as evidenced by:

- *Individual Assignment # 1 (20%) – due Friday, Sept. 12th*
- *Individual Assignment # 2 (20%) – due Friday, Oct. 3rd*
- *Project (40%) – due Friday, Oct. 10th*
- *Kahoot! (20%) – best 2 out of 3, 10% each*

Refer to the individual assignment and project briefings for details.

HOW TO BE SUCCESSFUL IN THIS COURSE

To succeed in this course, focus on:

1. ***Effective Communication:*** Students can create visualizations that clearly communicate the intended message to the target audience.
2. ***Appropriate Chart Selection:*** Students demonstrate the ability to choose the most effective graph or chart type to represent their data and support their message.
3. ***Clear and Impactful Visuals:*** Students create visualizations that are clear, engaging, and effectively convey insights from the data.

AI GUIDELINES

The use of artificial intelligence (AI) tools is not only permitted but also strongly encouraged in this course. Students are expected to explore various AI tools and technologies to enhance their learning experience and develop essential skills.

As you work with different AI models, focus on cultivating your critical thinking skills to effectively evaluate and assess the outputs, identifying both the benefits and limitations of AI-generated content. By embracing AI as a collaborative tool, you'll be better equipped to harness its potential and make informed decisions in your academic and professional pursuits.

GRADING & GRADING SCALE

Grading Scale

University Regulations on [Grades and Grade Reports](#) state that: Each student shall receive from their instructors a grade in each course for which they are enrolled at the close of the session. This grade shall indicate the student's achievement with respect to the objectives of the course.

In this class, grades reflect the sum of your achievements throughout the semester. You will accumulate points as described in the assignments portion above, with each assignment graded according to the rubric. At the end of the semester, final grades will be calculated by adding the total points earned and translating those numbers into the following letters (there will be no partial points or rounding). The exact grading scale will be determined at the end of the semester but use the following as a guideline throughout the course. Any alterations will be in favor of more generosity, not less.

- A 93-100%
- A- 90-92.99%
- B+ 87-89.99%
- B 83-86.99%
- B- 80-82.99%
- C+ 77-79.99%
- C 73-76.99%
- C- 70-72.99%
- D+ 67-69.99%
- D 63-66.99%
- D- 60-62.99%
- F 0-59.99%

The final grades will be curved. Per the School of Business Undergraduate Grading Policy, the target grade distribution for elective courses culminates in an average GPA that is around or below 3.3. The following table illustrates the target distribution.

Grade	A/A-	B+/B	B-	C+	C or below
GPA	4/3.7	3.3/3	2.7	2.4	<2
% of Students	30-40%	30-40%	10-20%	10-15%	< 10%

COURSE POLICIES

See Brightspace for details. Policies in Brightspace include the *Mental Health/Wellness, Accessibility, Attendance and Grief Absence Policy for Students, Office of the Dean of Students: Class Absences, Purdue Honor Pledge video, Purdue's Guide for Academic Integrity, Nondiscrimination Policy Statement, Violent Behavior Policy, and Use of Copyrighted Materials.*

SEMESTER OUTLINE

Due dates are subject to change if necessary. Any changes will be announced in class and updated on the calendar in Brightspace.

IMPORTANT! *In the event of a major campus or global emergency, course requirements, deadlines, and grading percentages are subject to change. To get information about changes in this course, check Brightspace or contact your instructor (email and phone are on the front page of this syllabus).*

Here's a link to the Purdue [Academic Calendar](#) and key University dates for the semester.

Module	Class Topic and/or Reading	Assignment
1 – Aug 25	Foundations of Data Visualization	
2 – Sept 1	Choosing the Right Chart	
3 – Sept 8	Design Principles for Visualization	Assignment 1 due Sept 12
4 – Sept 15	Storytelling with Data	
5 – Sept 22	Effective Visualization for Different Data Types	
6 – Sept 29	Iteration and Refinement	Assignment 2 due Oct 3
7 – Oct 6	Data Visualization in Practice Class Presentations	Project due Oct 10

* Schedule and assignments subject to change. Any changes will be posted in the learning management system.